

Fine-Grained Analysis of the Hyperscan Engine

Internship Project



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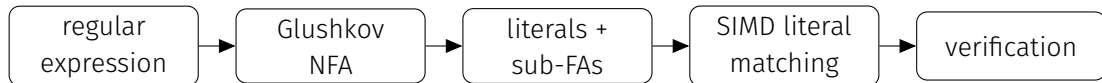
Motivation

Why Hyperscan?

- Modern algorithms cannot be fully understood by classical textbook techniques alone.
- In particular, modern regular expression engines are not just automata evaluators.
- Hyperscan combines graph decompositions, SIMD ¹ literal matchers (filtering and verification), and regular expression verifiers.
- This gives a productive source of theoretical questions.

¹Single Instruction, Multiple Data

From an Engine to a Statistic



Example: For $ab[0-9]^*xy$, Hyperscan finds positions of ab and xy , then verifies the middle part with $[0-9]^*$.

Guiding idea: Extracted literals are implementation details. Their lengths control how selective SIMD filtering can be.

Problem

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star ($*$).
- If $n \geq 3$, then the root is chosen as
 - concatenation ($.$) with probability u ,
 - alternation ($/$) with probability v ,
 - Kleene star ($*$) with probability $1 - u - v$.

Maximum-likelihood fitting on Snort rules² gives approximately $u \simeq 0.9$ and $v \simeq 0.09$.

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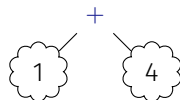
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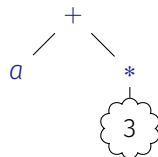
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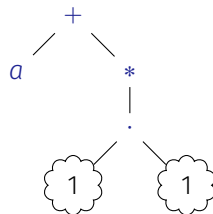
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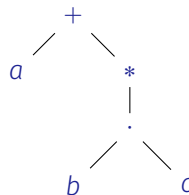
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The Longest-Literal Recurrence

The statistic of interest is

Y_n = length of the longest literal extracted from a random tree of size n .

Base cases:

- $Y_1 = 1$: a letter
- $Y_2 = 0$: $\star$ -rooted tree is ignored, since its language contains the empty string.

Let U_n be uniform on $\{1, \dots, n\}$. Let I and J be indicators of expectations u and v such that $I + J \leq 1$. Superscripts are used to denote independent copies.

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \underbrace{\left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right)}_{\text{take all of the subtrees}} + J^{(n)} \underbrace{\max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right)}_{\text{take the longer of the subtrees}}, \quad n \geq 1.$$

Problem: Analyze the asymptotic behavior of $\mathbb{E}[Y_n]$. In particular, is it true that $\mathbb{E}[Y_n] \sim Cn^\alpha$ for some $C, \alpha > 0$?

Deterministic Lower Bound

Bounding the Expectation

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), \quad n \geq 1, \quad Y_1 = 1, Y_2 = 0.$$

Take the expectation of both sides. The difficulty is the term

$$\mathbb{E} \left[\max \left(Y_i^{(1)}, Y_{n+1-i}^{(2)} \right) \right].$$

Using

$$\mathbb{E}[\max(X, Y)] \geq \max(\mathbb{E}[X], \mathbb{E}[Y]),$$

we get a deterministic lower-bound recurrence:

$$z_{n+2} = \frac{2u}{n} \sum_{i=1}^n z_i + \frac{v}{n} \sum_{i=1}^n \max(z_i, z_{n+1-i}), \quad z_1 = 1, z_2 = 0.$$

Main Deterministic Result

A power-law ansatz $z_n \sim Cn^\alpha$ gives

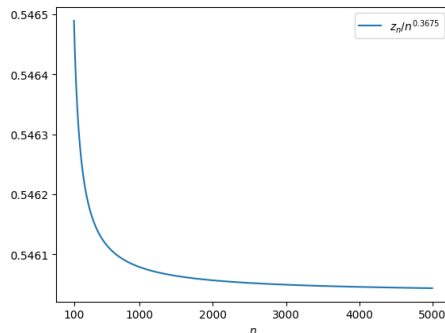
$$\alpha + v2^{-\alpha} = 2(u + v) - 1.$$

Theorem 1 (Exact asymptotics)

If $u > 1/2$, then

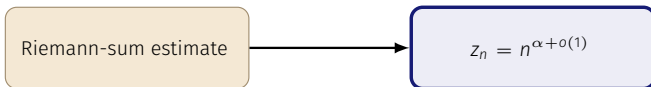
$$z_n \sim Cn^\alpha$$

for some $C > 0$, where α solves the equation above.

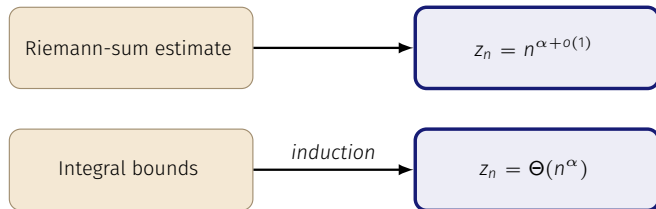


We take the ansatz $z_n \sim Cn^\alpha$ to find out intermediate results that must be true.

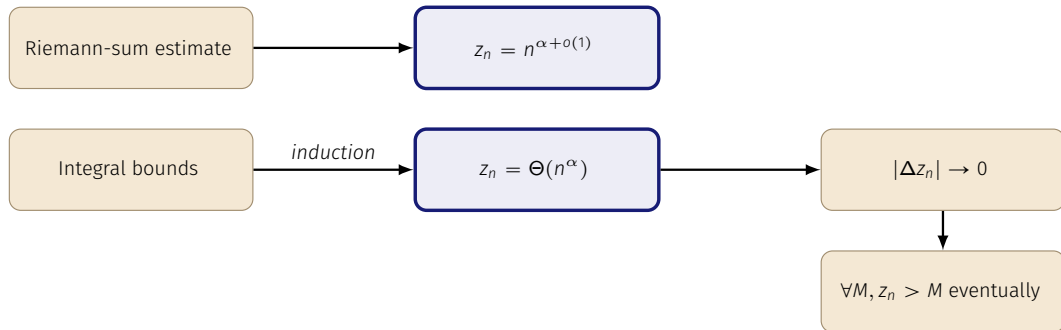
Proof Skeleton



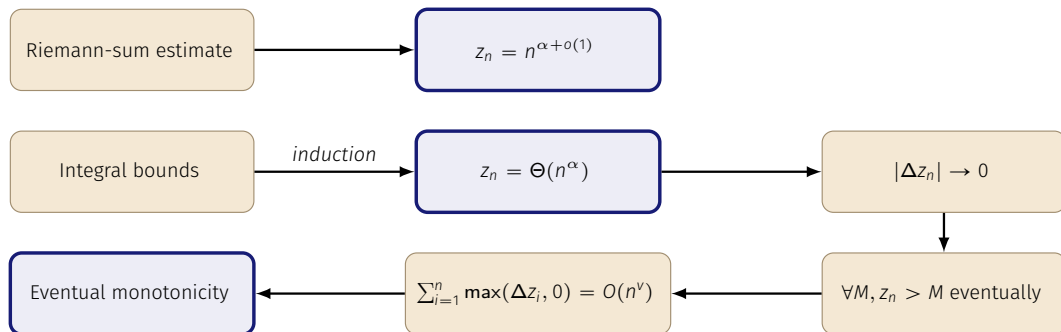
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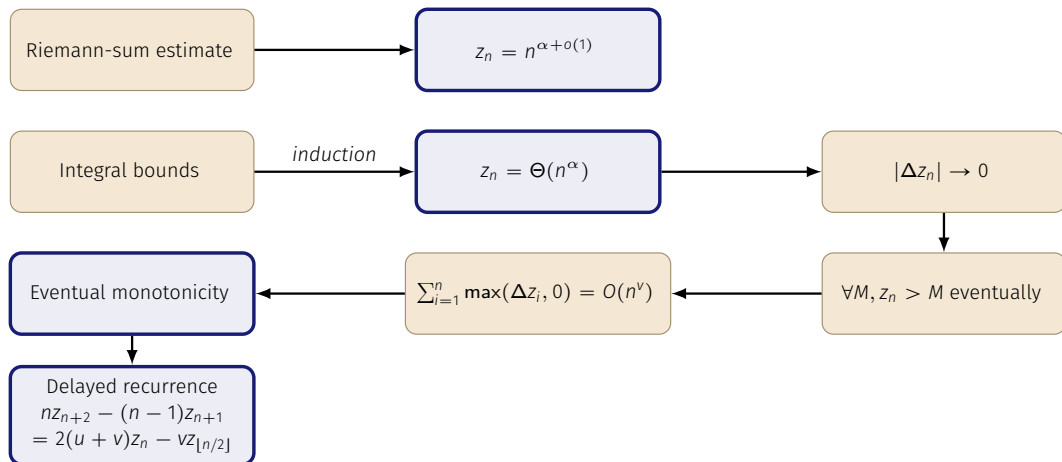
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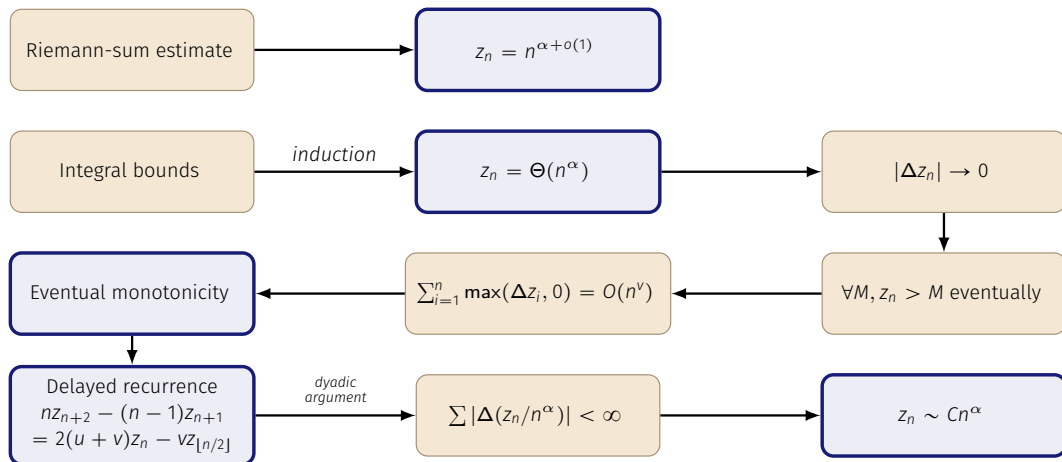
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Proof Skeleton



Stochastic Outlook

Back to the Stochastic Recurrence

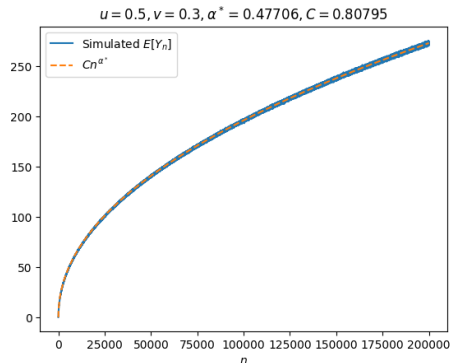
Simulations suggest that

$$y_n = \mathbb{E}[Y_n] \approx Cn^{\alpha^*}.$$

The exponent is larger than the deterministic lower-bound exponent because

$$\mathbb{E}[\max(X, Y)] > \max(\mathbb{E}[X], \mathbb{E}[Y])$$

in genuinely random cases.



A Fixed-Point Route

Normalize $X_n = Y_n/n^\alpha$. Any limiting distribution should satisfy

$$X \stackrel{d}{=} I \left(U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)} \right) + J \max \left(U^\alpha X^{(1)}, (1 - U)^\alpha X^{(2)} \right).$$

The direct contraction method ³ is obstructed by the fixed point δ_0 and by the fact that the operator does not preserve the mean.

The workaround is a mean-preserving iteration:

$$\mu_{n+1} = T_{\alpha_n}(\mu_n), \quad \mathbb{E}_{X \sim T_{\alpha_n}(\mu_n)}[X] = 1.$$

³Neininger, R., & Rüschendorf, L. (2005). Analysis of algorithms by the contraction method: additive and max-recursive sequences.

Existence via Tychonoff

- $\mathcal{M}(\mathbb{R})$: the space of bounded Radon measures on $\mathbb{R}$
- $\mathcal{D}(1)$: probability measures on $\mathbb{R}_+$ with mean 1

Mean-correcting exponent
choose $\alpha(\mu) \in (0, 1]$
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↓ *Prokhorov's theorem*

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Φ is continuous wrt.
the Wasserstein-1 metric

↓ *uniform integrability*

Φ is continuous wrt.
the weak topology

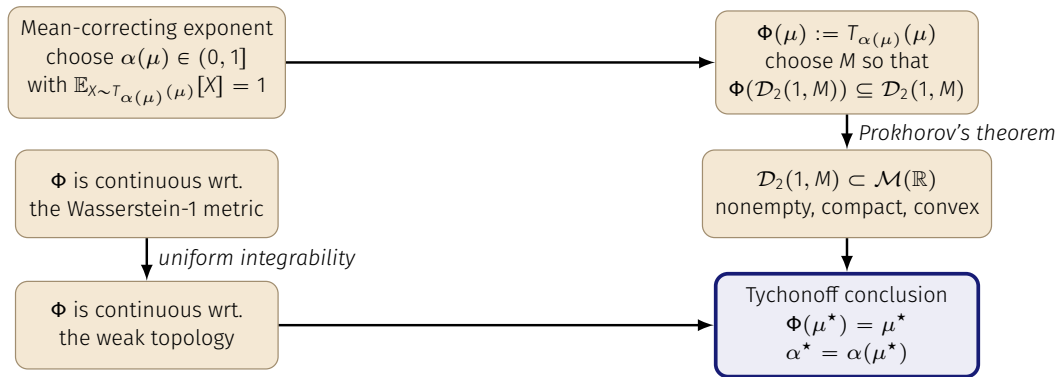
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- Hyperscan gives practical motivation for studying statistics of random regular expressions.
- The longest-literal statistic leads to a recurrence with sums, maxima, and resets.
- The deterministic lower bound has exact power-law asymptotics.
- The stochastic recurrence appears to have its own exponent α^* .

Main open direction

- Uniqueness of the fixed-point distribution μ^* .
- The asymptotics $y_n = n^{\alpha^* + o(1)}$, characterization α^* .
- Convergence of the mean-preserving fixed-point iteration.